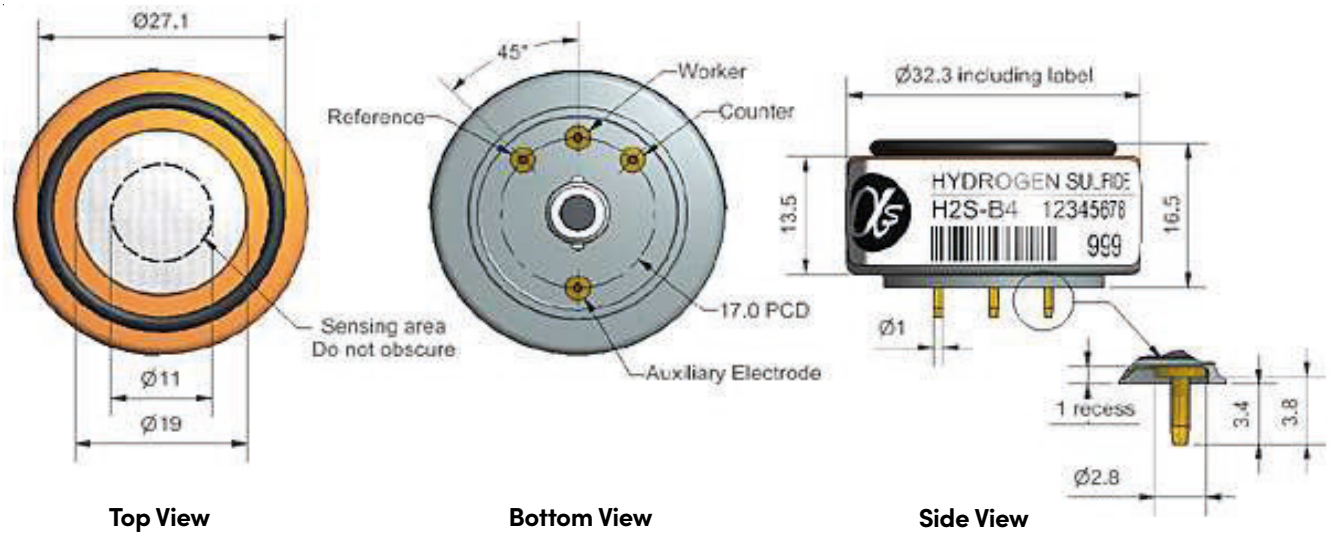
H2S-B4 Hydrogen Sulfide Sensor – 4-Electrode



Top View **Bottom View** **Side View**

Dimensions are in millimetres (± 0.15 mm).

Performance	Sensitivity	nA/ppm at 2ppm H_2S		1450 to 2600
	Response time	t90 (s) from zero to 2ppm H_2S		< 60
	Zero current	nA in zero air at 20°C		-250 to 200
	Noise*	±2 standard deviations (ppb equivalent)		1
	Range	ppm H_2S limit of performance warranty		100
	Linearity	ppb error at full scale, linear at zero and 40ppm H_2S		< ± 4
	Overgas limit	maximum ppm for stable response to gas pulse		200
	***Tested with Alphasense ISB low noise circuit**			
Lifetime	Zero drift	ppm equivalent change/year in lab air		< ± 100
	Sensitivity drift	% change/year in lab air, monthly test		< 20
	Operating life	months until 50% original signal (24-month warranted)		> 24
Environmental	Sensitivity @ -20°C	% (output @ -20°C/output @ 20°C) @ 2ppm H_2S		77 to 90
	Sensitivity @ 50°C	% (output @ 50°C/output @ 20°C) @ 2ppm H_2S		100 to 110
	Zero @ -20°C	nA change from 20°C		50 to 60
	Zero @ 50°C	nA change from 20°C		-120 to -160
Cross Sensitivity	NO_2 sensitivity	% measured gas @ 5ppm	NO_2	< -10
	Cl_2 sensitivity	% measured gas @ 5ppm	Cl_2	< -12
	NO sensitivity	% measured gas @ 5ppm	NO	< 12
	SO_2 sensitivity	% measured gas @ 5ppm	SO_2	< 20
	CO sensitivity	% measured gas @ 5ppm	CO	< 3
	H_2 sensitivity	% measured gas @ 100ppm	H_2	< 0.5
	C_2H_4 sensitivity	% measured gas @ 100ppm	C_2H_4	< 0.1
	NH_3 sensitivity	% measured gas @ 20ppm	NH_3	< 0.1
	CO_2 sensitivity	% measured gas @ 5%	CO_2	< 0.1
Key Specifications	Temperature range	°C		-30 to 50
	Pressure range	kPa		80 to 120
	Humidity range	% rh continuous		15 to 90
	Storage period	months @ 3 to 20°C (stored in sealed pot)		6
	Load resistor	Ω (ISB circuit is recommended)		33 to 100
	Weight	g		< 13

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For further information on the performance of this sensor, on other sensors in the range or any other subject, please contact Alphasense Ltd. or visit our website at "www.alphasense.com".

Figure 1 Sensitivity Temperature Dependence

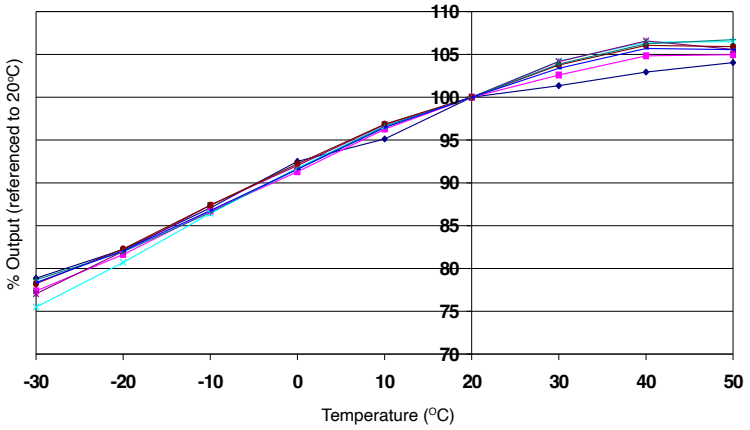

Figure 1 shows the temperature dependence of sensitivity at 2ppm H_2S.

This data is taken from a typical batch of sensors.

Figure 2 Zero Temperature Dependence

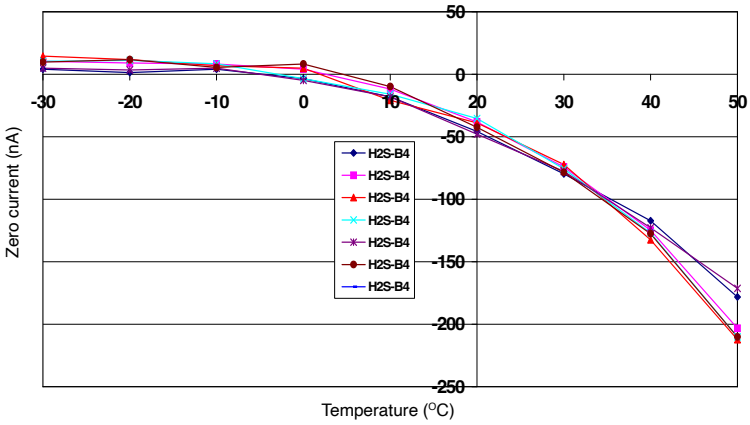

Figure 2 shows the variation in zero output of the working electrode caused by changes in temperature, expressed as nA.

This data is taken from a typical batch of sensors.

Contact Alphasense for futher information on zero current correction.

Figure 3 Linearity to 200 ppb H_2S

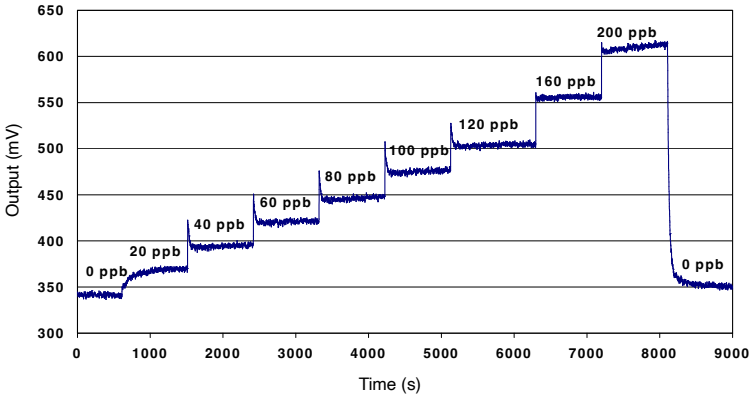

Figure 3 shows response to 200ppb H_2S.

Use of Alphasense ISB circuit reduces noise to 1ppb, with the opportunity of digital smoothing to reduce noise even further.

At the end of the product's life, do not dispose of any electronic sensor, component or instrument in the domestic waste, but contact the instrument manufacturer, Alphasense or its distributor for disposal instructions. NOTE: all sensors are tested at ambient environmental conditions unless otherwise stated. As applications of use are outside our control, the information provided is given without legal responsibility. Customers should test under their own conditions, to ensure that the sensors are suitable for their own requirements.

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